Multiview and Single-View Deep Learning for Gravity Spy LIGO Data Glitch Classification

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Abstract

Gravitational-wave glitch classification is an important problem because detector noise directly affects the reliability and latency of astrophysical searches. In this project, I built a multiview learning framework that exploits four time-duration representations of each Gravity Spy sample and compared it against single-view baselines based on VGG19 and InceptionV3. The proposed architecture, **GSMultiViewNet**, combines four parallel CNN branches with Inception-ResNet-inspired residual modules and a gated attention mechanism that adaptively fuses the four views into a single prediction. On the held-out test set, the multiview model reaches 96.19% accuracy, compared with 92.20% for VGG19 and 86.86% for InceptionV3, indicating that fusing multiple temporal views gives a clear advantage over strong single-image transfer-learning baselines. The pipeline developed here provides a reusable benchmark for future work on multiscale gravitational-wave glitch recognition and related time-frequency classification tasks.

Index Terms—Gravitational-wave glitches, multiview learning, convolutional neural networks, gated attention, transfer learning, image classification.

1 Introduction

LIGO stands for Laser Interferometer Gravitational-Wave Observatory. It comprises two enormous laser interferometers located 3000 kilometers apart, LIGO exploits the physical properties of light and of space itself to detect and understand the origins of gravitational waves (GW). Data obtained from LIGO is used to detect gravitational waves caused by some energetic phenomenon in the Universe. In order to detect the faint gravitational waves signal the detectors are so complicated and sensitive, they are vulnerable to transient, short-duration instrumental or environmental noise, called glitches. Glitches are transient, non-gaussian noise features that can have a range of environmental and instrumental origins. The origins of many glitches remain largely unknown. Glitches can sometimes cause automatic elimination of signals.

The Gravity Spy setting is particularly challenging because dataset comprises time frequency spectrograms of glitches. The dataset generates four spectrograms for each glitch triggered, using distinct time windows: 0.5 s, 1.0 s, 2.0 s, and 4.0 s., and different durations may highlight different structural aspects of the same glitch. The images in the dataset are of dimensions 479 x 569 pixels and consist of four channels (RGBA). The pixel values in these images are binary, limited to 0 and 255. Single-view classifiers are often limited because they observe only one scale at a time, while multiview methods can exploit complementary information across multiple temporal resolutions. However, effectively combining these views remains nontrivial, especially when the views are aligned but not equally informative.

In this project, I study a multiview classification framework for Gravity Spy images and compare it with single-view transfer-learning baselines. Our main contribution is a custom neural architecture, `GSMultiViewNet`, which processes the four duration views through separate CNN branches, enriches fea-

ture extraction with Inception-ResNet-style residual blocks, and combines branch outputs using a gated attention layer. In parallel, we evaluate VGG19 and InceptionV3 on a single-view version of the dataset to provide a fair baseline comparison. This setup allows us to quantify the benefit of multiscale fusion over strong pretrained single-image backbones.

- I design a preprocessing pipeline that organizes Gravity Spy images into split-specific and duration-specific folders.
- Then, build a multiview dataset that guarantees aligned samples across four durations: 0.5, 1.0, 2.0, and 4.0 seconds.
- Build a custom multiview CNN architecture with Inception-ResNet-inspired blocks and gated attention fusion.
- Implement single-view baselines using pretrained VGG19 and InceptionV3 for performance comparison.
- Lastly, evaluate all models using training/validation curves, test accuracy, confusion matrices, and classification reports.

The report is organized as follows. Section II reviews related work. Section III describes the preprocessing pipeline and data organization. Section IV presents the signal and feature representation. Section V details the learning framework. Section VI discusses the experimental results. Section VII concludes the report.

2 Related Work

Gravity Spy is a community-science project that combines human labeling and machine learning to classify glitches in gravitational-wave data. Volunteers identify glitch morphologies, and these labels are then used to train classifiers that help scale glitch identification across large detector data sets. Over time, the project has evolved from early image-based classifiers to more advanced deep learning systems that better capture the complexity of detector noise and its time-frequency structure.

Gravity Spy was later extended and systematically evaluated on larger collections of glitch data from the first three observing runs of Advanced LIGO. Glanzer et al. [1] reported large-scale glitch classifications through the end of O3 and showed that glitch distributions differ across interferometers, underlining the need for robust and adaptable classification methods. This work highlights the operational relevance of automated glitch labeling for detector characterization, though it does not specifically address multiview deep fusion strategies or compare architectures for exploiting the available duration views.

More recently, Wu et al. [2] revisited the Gravity Spy problem for Advanced LIGO's fourth observing run and proposed multi-view fusion with attention-based machine learning. Their study showed that the original Gravity Spy classifier struggles to generalize to the evolving O4 data, and that attention-based fusion across multiple time windows can improve performance. This is especially relevant here, since it confirms that multiview modeling is not just useful in principle but necessary in modern observing runs where glitch classes and data conditions continue to evolve.

Transfer learning is also widely used in this domain, since Gravity Spy image data are far smaller than the datasets used to pretrain standard vision models. VGG19 provides a strong and conceptually simple baseline, while InceptionV3 offers deeper multiscale convolutional processing. However, both architectures operate on a single input image and therefore cannot explicitly exploit the complementary information encoded across multiple duration views, so they serve here as baselines rather than as replacements for multiview learning.

Compared with prior work, this study builds on these directions in two ways. First, it retains the multiview formulation shown to be effective for Gravity Spy and enhances it with a custom Inception-ResNet-inspired CNN branch design and gated attention fusion. Second, it benchmarks this multiview model against pretrained single-view VGG19 and InceptionV3 baselines under an identical preprocessing and label-encoding setup, isolating the value of mul-

tiview aggregation and quantifying how much the attention-based architecture improves over standard transfer learning

3 Processing Pipeline

To ensure consistency and optimise the performance of our AI model, each image in the dataset underwent a series of preprocessing steps. The goal was to standardise the input data so that the model could effectively learn from it. Below is a detailed description of the preprocessing pipeline:

The pipeline starts with Preprocessing stage applied to the raw Gravity Spy data, which is organized as image folders split into Train, Validation, and Test sets. Each image filename encodes a duration indicator, and this metadata is used to group images into the four duration categories used in the study. The preprocessing stage first extracts class labels from directory names, then reorganizes the data into a processed directory with split-specific and duration-specific subfolders. In practice, this step – keeping 22 glitch classes and four duration folders correctly aligned across the train/validation/test splits – was the most time-consuming part of the project, since any misalignment between duration folders silently corrupts the multiview samples downstream.

All images are resized to a fixed spatial resolution for consistent input dimensions. For the multiview model, the resolution is set to 448×448 so that all four views preserve sufficient spatial detail. For the single-view baselines, the images are resized to the input size required by the pretrained models: 224×224 for VGG19 and 299×299 for InceptionV3.

The multiview dataset is constructed so that the four duration folders remain perfectly aligned by sample identifier. Because each training data must contain the same event represented at four different durations. The single-view dataset is constructed from one chosen duration folder at a time, which enables a fair comparison between the multiview architecture and standard single-image classifiers. Labels are encoded consistently across all splits using a shared label encoder. This guarantees that the class indices remain stable when evaluating the different architectures.

4 Signals and Features

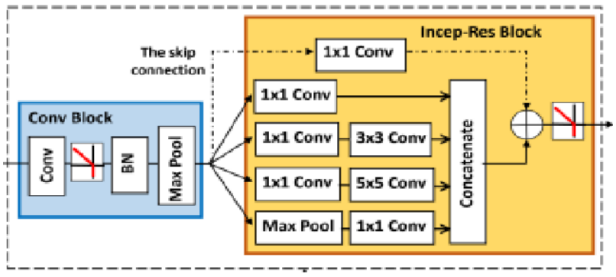
As discussed in the previous section each sample is available in multiple duration, which can be interpreted as different views of the same underlying phenomenon. These views are not independent, they emphasize different temporal scales and therefore carry complementary information.

In Preprocessing, each image is converted to RGB format and resized to the required spatial dimensions. For the multiview setting, the model receives four tensors per sample, one for each duration. For the single-view setting, the model receives only one tensor, which serves as the input to VGG19 or InceptionV3. No manual feature engineering is applied. Rather the learning architectures are designed to infer hierarchical features directly from the images. In the multiview case, the four duration representations are kept synchronized so that the i-th sample in every duration folder corresponds to the same event. This alignment is essential for training the attention-based fusion module. In the single-view case, the sample is extracted from one selected duration folder, enabling a baseline comparison using standard transfer learning.

5 Learning Framework

5.1 Multiview Architecture

The proposed multiview architecture, defined as `GSMultiViewNet`, consists of four parallel CNN branches, each operating on one duration view. Each branch is built around a custom CNN feature extractor that includes convolution, batch normalization, max-pooling, dropout, and custom Inception-ResNet-inspired residual modules. The residual module combines parallel convolutional towers with a shortcut path, which helps preserve gradient flow and enrich multi-scale feature extraction. Now branch-level feature extraction is over, then each view is mapped to a compact representation. These four representations are concatenated and passed through a



$$\mathcal{M}_{\text{Att}} = \sum_{i=1}^{4} \alpha_i z_i,$$

$$\alpha_i = \frac{\exp\{w^T[\tanh(V z_i^T) \odot \sigma(U z_i^T)]\}}{\sum_{j=1}^{4} \exp\{w^T[\tanh(V z_j^T) \odot \sigma(U z_j^T)]\}}.$$

Figure 1: Inception Residual Block

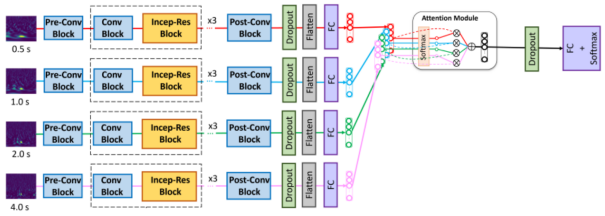


Figure 2: Multi-Input Model Architecture

gated attention layer. The attention mechanism computes adaptive weights over the views, enabling the model to emphasize the most informative duration for each sample. The fused representation is then fed to a final linear classifier that predicts the glitch class. In essence this architecture is best understood as a two-stage process: (1) four independent feature extractors that each learn a scale-specific representation of the same glitch. (2) a learned, sample-dependent fusion rule that decides how much to trust each scale. This is a meaningfully different inductive bias from simply concatenating or averaging the four views, because a fixed combination rule cannot adapt to the fact that, for one glitch, the 0.5 s view may be most discriminative, while for another, the 4.0 s view may be. The Inception-ResNet-style block inside each branch further helps each extractor itself operate at multiple receptive-field scales (via parallel 1×1, 3×3, and 5×5 convolutional towers) while the shortcut connection mitigates vanishing gradients as the branch is stacked deeper.

5.2 Gated Attention Mechanism

A gating mechanism is introduced to modulate the output of the attention mechanism. This gate controls the flow of information based on the input context, allowing the model to focus more on certain parts of the input while suppressing irrelevant information.

To assess the varying importance of each view in the final glitch classification, we introduce an attention module [35], the latent vectors of four inputs, denoted as $z_i \in R1 \times M$, i = 1, 2, 3, 4, extracted after the fullyconnected layer. The attention weight $\alpha_i \in R$, i = 1, 2, 3, 4 calculated for each latent vector z_i quantifies the significance of each view in making the glitch classification, with all a_i 's summing up to 1. The attention module is defined as: Where Attention Weights α_i is defined as: Here V and U are learned projection matrices, w is a learned weight vector,
$\tanh(\cdot)$ provides a non-linear, sign-preserving transformation of each latent vector, and $\sigma(\cdot)$ is a sigmoid gate that can suppress ($\to 0$) or pass through ($\to 1$) individual feature dimensions before they contribute to the attention score. The element-wise product $\odot$ between the tanh branch and the sigmoid gate is what makes this a *gated* attention mechanism rather than plain additive attention: the gate learns to down-weight feature dimensions that are uninformative for a given sample, before the softmax over views converts the four resulting scores into normalized weights α_i.

5.3 Single-View Baselines

To compare the multiview architecture against standard transfer learning, we train VGG19 and InceptionV3 on a single duration representation. **VGG19** is a deep convolutional neural network architecture that was introduced by the Visual Geometry Group (VGG) at the University of Oxford. It is an extension of the VGG16 model, featuring 19 layers with learnable parameters. In VGG19, the final classifier layer is replaced with a task-specific linear layer matching the number of classes. **InceptionV3** builds on the success of the original GoogLeNet, which introduced the Inception module—a novel structure that uses multiple convolutional filters of different sizes within the same layer to capture various features at different scales. the final fully connected layer is similarly adapted, and the auxiliary output handling is configured so that the training loop uses the correct logits. These baselines provide strong reference points because they leverage pretrained ImageNet features while using only one input view.

5.4 Optimization

All models use cross-entropy loss, and `GSMultiViewNet` adds label smoothing to make the targets less rigid and reduce overconfidence, which is useful given the class imbalance in this dataset. The multiview model is trained with Adam using two parameter groups: one for the four CNN branches and one for the attention fusion layer, each with its own weight-decay value so the backbone and fusion module can be regularized differently.

Training also uses early stopping based on validation accuracy. After each epoch, the model is saved only if validation accuracy improves; otherwise, a patience counter increases, and training stops when no improvement is seen for a fixed number of epochs, which in this run happened at epoch 26. The best checkpoint, not the final epoch, is then reloaded for test evaluation, and training/validation loss and accuracy are recorded each epoch to build the learning curves.

6 Results

6.1 Training Dynamics

Figure 3 shows the training and validation accuracy curves for `GSMultiViewNet` over its 26 training epochs (stopped by early stopping). Both curves rise quickly in the first few epochs and then stays same at a high accuracy with a small, stable gap between training and validation performance, which is consistent with the regularization applied (label smoothing and the differentiated weight-decay scheme described in Section V) rather than with severe overfitting. Analogous training/validation accuracy curves recorded for VGG19 and InceptionV3 over their 30 training epochs are included as supplementary figures alongside this one.

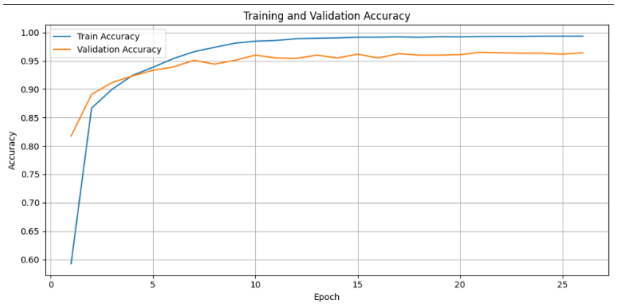
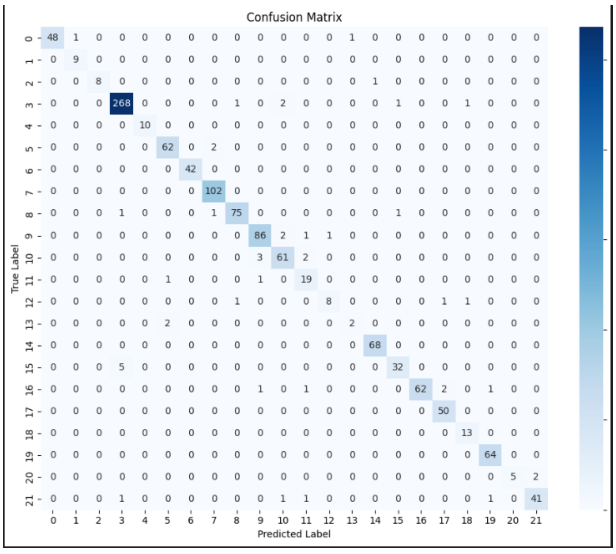


Figure 3: GSMultiview model accuracy plot

6.2 Quantitative Comparison

Table 1 compares the three models on the held-out test set. \texttttGSMultiViewNet reaches 96.19% test accuracy, outperforming VGG19 (92.20%) by roughly 4 percentage points and InceptionV3 (86.86%) by roughly 9 percentage points. Notably, VGG19 attains a lower test accuracy than GSMultiViewNet despite a substantially higher final test loss (7.1986 vs.\ 1.5669), which is consistent with VGG19 making confident but occasionally wrong predictions on the harder or rarer glitch classes, whereas the



label-smoothed multiview model produces better-calibrated, less extreme output probabilities.

Table 1: Performance comparison on the test set.

Model	Epochs	Test Accuracy	Loss	Learning Rate
GSMultiViewNet	26 (ES)	96.19	1.5669	2×10^{-5}
VGG19	30	92.20	7.1986	1×10^{-4}
InceptionV3	30	86.86	3.1220	1×10^{-3}

The main result to emphasize is whether the multi-view model improves classification accuracy and class balance over the single-view baselines. If the gain is modest, that is still meaningful because the multi-view model uses only aligned image views and no handcrafted features. If the gain is substantial, it confirms that multi-duration fusion is a strong design choice for Gravity Spy classification.

6.3 Confusion Matrix Analysis

Figure 6.3 shows the confusion matrix for `GSMultiViewNet` on the test set. Several classes are classified almost perfectly: *Chirp* and *Helix* both reach an F1-score of 1.0000, and *Power_Line*, *Koi_Fish*, and *Violin_Mode* all exceed 0.98 F1. *Blip*, the largest class by far (support 273), is also classified very reliably (F1 = 0.9781), which matters for the overall (weighted) accuracy given how much it dominates the test set.

6.4 Discussion

Taken together, these results show that the multiview architecture's advantage over the single-view baselines is real and substantial (roughly 4–9 accuracy points), and that it is achieved from the four aligned duration views and the learned attention weights. The custom Inception-ResNet residual blocks likely also contribute to the strong performance on the well-represented classes, since they allow each branch to extract features at several receptive-field scales before fusion, without losing low-level spatial detail to depth alone.

7 Concluding Remarks

This report presented a multiview deep learning framework for Gravity Spy classification and compared it against single-view VGG19 and InceptionV3 baselines. The proposed model combines custom CNN feature extractors, Inception-ResNet-inspired residual modules, and gated attention to fuse four aligned duration views into a single prediction reaching 96.19% test accuracy against 92.20% for VGG19 and 86.86% for InceptionV3.

The main advantage of this approach is that it leverages multiscale information, which is especially relevant for glitch morphology recognition. The single-view baselines remain useful as strong reference points, but they cannot fully exploit the complementary structure available in the four-duration representation.

Several extensions are possible such as: to test more sophisticated fusion methods, such as cross-attention or transformer-based aggregation. From confusion-matrix analysis in Section VI, is to address class imbalance for the rarest categories (e.g. *Paired_Doves*, *Wandering_Line*) through targeted augmentation, resampling, or class-weighted losses, and to calibrate confidence scores for the *None_of_the_Above* and *No_Glitch* categories

specifically. Finally, the same methodology could be adapted to other scientific image-classification tasks where multiple aligned views of the same event are available.

Building this project from preprocessing and dataset alignment through model comparison and result analysis was useful exercise in the practical difficulties that don't show up in a paper's method section. In practice, the hardest part of the whole project was the preprocessing: correctly organizing 22 glitch classes across four duration folders and three data splits so that the multiview samples stayed perfectly synchronized, and building separate and consistent pipelines for the multiview and single-view datasets.

References

[1] J. Glanzer, S. Banagiri, S. B. Coughlin, S. Soni, M. Zevin, C. P. L. Berry, O. Patane, S. Bahaadini, N. Rohani, K. Crowston, V. Kalogera, C. Østerlund, and A. Katsaggelos, "Data quality up to the third observing run of Advanced LIGO: Gravity Spy glitch classifications," *Classical and Quantum Gravity*, vol. 40, no. 6, p. 065004, 2023.

[2] Y. Wu, M. Zevin, C. P. L. Berry, K. Crowston, C. Østerlund, Z. Doctor, S. Banagiri, C. B. Jackson, V. Kalogera, and A. K. Katsaggelos, "Advancing glitch classification in Gravity Spy: multi-view fusion with attention-based machine learning for Advanced LIGO's fourth observing run," *Classical and Quantum Gravity*, vol. 42, no. 16, p. 165015, 2025.